CI Compass FAIR Topical Working Group

Experience with data licensing and reuse

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NSF major and mid-scale facilities generally aim to make their data publicly accessible, reusable, and well documented. Part of this is using inclusive licenses that allow widespread use. Each facility approaches this in a unique way, and licensing terms are generally not facility wide, but instead commonly defined at the dataset, program, repository, or contributing-network level. Several facilities rely primarily on access policies, attribution requirements, or repository-specific terms. However, where an explicit license is provided, it is usually a permissive Creative Commons or public-domain framework license, such as CC-BY-4.0 or CC0.

A recent concern is the rapid growth in data-download volumes and associated infrastructure and egress costs, particularly for cloud-hosted data. A significant portion of this growth has been attributed to automated scraping and bot traffic, including harvesting for AI-related activities. There are several methods that have been explored or enacted by facilities:

- Making licensing and attribution requirements more explicit, or moving to slightly less permissive open licenses, such as from CC0 to CC BY. This makes the attribution of the data source an explicit requirement.

- Adding extra download steps to reduce uncontrolled automated harvesting. This includes requiring authentication for data access, even through free user accounts. This contrasts with previously allowing unlimited and free data downloads from unauthenticated users.

- Exploring alternative access models such as the Open Science Data Fabric (OSDF), which preserves open access while requiring users and workflows to use a managed data-transfer system rather than anonymous HTTP downloads.

- Exploring ways for AI agents to use facility resources more efficiently, such as the provision of skills documents for AI agents to discover and use, which will improve their efficiency in accessing data. And the use of cloud-native files that allow for exploring datasets without downloading, and also data subsetting prior to download.

- Use persistent identifiers for licenses where possible, e.g. Creative Commons URLs or SPDX identifiers.

It remains to be seen what effects these changes will have on the research community, in terms of data discovery and accessibility, or whether AI use by researchers will increasingly strain the financial and resource sustainability of facilities.

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